# HAMDARD UNIVERSITY BANGLADESH

**Faculty of Science, Engineering and Technology**

**Department of Computer Science and Engineering**

**CSE 415: Communication Engineering**

Batch : 23rd
Date: 03/03/2026

Final Examination

Semester: Fall 2025
Time: 2 Hours
Total Marks : 40

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Answer any **five** questions: $5 \times 8 = 40$

1. (a) In ATM, what is the relationship among TPs, VPs, and VCs? 4
   (b) Briefly explain the ATM design goals. 4

2. (a) State the relationship between STS and STM. 4
   (b) Describe the functions of each SONET layer. 4

3. (a) Describe the SS7 service and its relation to the telephone network. 4
   (b) How data transfer is achieved using CATV channels? Explain. 4

4. (a) Compare a Piconet and a Scatternet in the Bluetooth architecture. 4
   (b) Write the role of the L2CAP layer in Bluetooth. 4

5. (a) Explain the differences between fixed WiMAX and mobile WiMAX. 4
   (b) What are the functions of a mobile switching center? 4

6. (a) Illustrate IS-95 reverse transmission. 4
   (b) Write the relationship between D-AMPS and AMPS. 4

7. (a) What are the purposes of GPS? Explain. 4
   (b) Which is better: a low frequency reuse factor or a high reuse factor? Explain your answer. 4

# HAMDARD UNIVERSITY BANGLADESH

**Faculty of Science, Engineering and Technology**
**Department of Computer Science and Engineering**
**CSE413: Computer Graphics**

Batch: 23rd
Semester: Fall 2025

Date: 01/03/2026
**Final Examination**
Time: 2 hours
Total Marks: 40

Answer any **five** questions from the following: 5x8= 40

| | | | |
|---|---|---|---|
| 1 | (a) | State the properties of ellipse. | 2 |
| | (b) | Using the midpoint ellipse algorithm, show how to calculate the incremental points in Region 1 of the ellipse. | 2 |
| | (c) | Consider an ellipse with a semi-major axis a=8 units and a semi-minor axis b=5 units. Use the midpoint ellipse algorithm to determine the initial points of the ellipse starting from (0, b) and progressing toward the first quadrant. | 4 |
| 2 | (a) | Describe the visible surface detection algorithms. | 3 |
| | (b) | Explain the Painter's Algorithm along with its advantages and disadvantages. | 5 |
| 3 | (a) | Explain the brute-force line clipping method. | 3 |
| | (b) | Determine whether the following lines are accepted/rejected/partial using Cohen Sutherland line clipping algorithm. Given the clipping region with corners (−200,−150) to (200,150): i. (−120,−180) to (220,−160) ii. (−200,150) to (200,−150) | 5 |
| 4 | (a) | Define color. | 2 |
| | (b) | Convert the RGB color R=90, G=180, B=210 to the HSL color model. Additionally, represent the same color in the CMY and YIQ color models. | 6 |
| 5 | (a) | State the differences between orthogonal and oblique projection. | 2 |
| | (b) | A 3D point P(25.0, 15.0, −600.0) is projected onto a projection plane whose center is at (0.0, 0.0, −400.0). The center of projection (COP) is located at (50.0, 30.0, 0.0). Determine the coordinates of the projected point on the projection plane using the general-purpose projection matrix approach. | 6 |
| 6 | (a) | Discuss the factors that make clipping operations difficult in computer graphics, providing examples for lines and polygons. | 3 |
| | (b) | A triangle has vertices P(2,2), Q(5,2), and R(3,5). Clip this triangle against a rectangular clipping window $x_{min}$=3, $y_{min}$=3, $x_{max}$=6, and $y_{max}$=4 using the Sutherland–Hodgman algorithm. List the clipped polygon vertices. | 5 |

Batch: 23rd



S → AAB
A → aA|C
B → jkl| tuv
C → xyz

Whether the above grammar is accepted by a CLR(1) parser or not. Write down why a CLR (1) parser is better than an SLR(1) parser or an LL(1) parser?

7. (a) S →AB 8
A → Aab|C
B → prs
C → xyz

Show the above grammar is accepted by LALR(1) parser or not, and also write down why a CLR(1) parser is not an LALR(1) parser?

# HAMDARD UNIVERSITY BANGLADESH

**Faculty of Science, Engineering and Technology**
**Department of Computer Science and Engineering**
**CSE 417: Compiler Design**

Batch: 23rd | Semester: Fall 2025

Date: 05/03/2026 | **Final Examination** | Time: 2 hours

Total Marks: 40

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Answer any five questions. **Marks: 5 * 8 = 40**

1. (a) Define Context-free grammar and Recursive grammar. 2
   (b) Write the difference between SDD and SDT. 3
   (c) State the summary of a recursive descent parsing in a compiler. 3

2. (a) Write down about Ambiguous grammar and Unambiguous grammar. 2
   (b) Write a C program to sum two numbers. Then generate a symbol table for the program. 3

   (c) $S \rightarrow AA$ $A \rightarrow aA/b$ 3
   Whether the above grammar is accepted by LL(1) parser or not.

3. (a) Write about an annotated tree. 2
   (b) Make the table for Quadruples for the expression : k = c* - b + c * - c +a 3
   (c) $S \rightarrow AAB$ 3
   $A \rightarrow aA|b$
   $B \rightarrow jkl|\ tuv$
   Show the above grammar is accepted by SLR(1) parser or not.

4. (a) State the differences between an annotated parse tree and a syntax tree. 2
   (b) Draw an annotated tree for : (9+8*(7+6))*4n . 6

5. (a) State the rules for Quadruples. 3

   (b) $S \rightarrow S \# A / A$ 5
   $A \rightarrow A \& B / B$
   $B \rightarrow id$
   Given String : 10 & 4 # 6 .
   Draw annotate tree from the given grammar and show the output for the given string.

6. (a) $E \rightarrow E + T$ 3
   $E \rightarrow T$
   $T \rightarrow T * F$
   $T \rightarrow F$
   $F \rightarrow id$
   Draw the dependency graph for: 10 * 4 + 3 * 7 + 8.
   (b) $S \rightarrow AAB$ 5

(a) Identify Support Vectors. 2
(b) Calculate the Decision Boundary. 4
(c) Compute Margin. 2

4. (a) Draw a neural network using 10 Neurons and 2 layers. 3
(b) Calculate the probability using sigmoid function for a neuron. Given that, w=2, b=-1, x=10. 2
(c) Define forward and backward training for a neural network. 3

5. Given that,

| Class | Predicted |
|---|---|
| 1 | 1 |
| 1 | 0 |
| 0 | 0 |
| 0 | 1 |
| 1 | 1 |
| 0 | 0 |

(a) Calculate the TP, TN, FN and FP. 3
(b) Draw the Confusion Matrix. 2
(c) Calculate the Accuracy, Sensitivity and Specificity. 3

6. *Given that, data = [1,2,3,4,5]*
*(a) Write a python function that returns the average.* 2
*(b) Write a python function that returns the Standard Deviation* 4
(c) Write a python function that returns the median. 2

7. Define True or False.
(a) Average is highly influenced by outliers. T 1
(b) Regression supports kernel trick. T 1
(c) Decision tree is prone to overfitting. T 1
(d) Random-forest is not an ensembled model. T 1
(e) Sigmoid function returns between values -1 to 1. F 1
(f) Specificity calculates the ability to predict true negatives of a model. T 1
(g) Sensitivity calculates the ability to predict true negatives of a model. F 1
(h) In DT, the leaf node makes a prediction based on majority vote. T 1

# HAMDARD UNIVERSITY BANGLADESH

**Faculty of Science, Engineering and Technology**
**Department of Computer Science and Engineering**
**CSE 411: Artificial Intelligence**

Batch: 23rd | Semester: Fall 2025

Time: 2 hours

Date: 08/03/2026 | **Final Examination** | Total Marks: 40

Answer any **Five** questions from the following. Marks: 5 x 8 = 40

1. Given that

| Study Hours (X) | Exam Score (Y) |
|---|---|
| 2 | 50 |
| 3 | 55 |
| 5 | 65 |
| 7 | 70 |
| 9 | 85 |

(a) Calculate the regression line using the equation: $Y = a + bX$, Where X is study hours and Y is exam score. 3

(b) Given that: $9 < \text{study hour} < 2$, Justify if it is interpolation or extrapolation and write your reason. 2

(c) Find the standard error of regression. 3

2. Given that,

| **Sample** | **Weather ($X_1$)** | **Temperature ($X_2$)** | **Play Tennis (Y)** |
|---|---|---|---|
| 1 | Sunny | Hot | No |
| 2 | Sunny | Cool | Yes |
| 3 | Rainy | Hot | Yes |
| 4 | Rainy | Cool | Yes |
| 5 | Sunny | Hot | No |

(a) Calculate the Total Entropy of Dependent Variable. 2

(b) Calculate the information gain of each feature split. 3

(c) Select the Root Node and Draw the Decision tree based on Information Gain. 3

3. Given That,

| Sample | $x_1$ | $x_2$ | Class (y) |
|---|---|---|---|
| A | 1 | 2 | -1 |
| B | 2 | 3 | -1 |
| C | 3 | 3 | +1 |
| D | 5 | 6 | +1 |
| E | 6 | 7 | +1 |

b) Explain why MIME and S/MIME protocols were introduced after SMTP. 4

7. Illustrate and explain the steps involved in the Kerberos authentication protocol when a client requests access to a local server within a network. 8

# HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

**Department of Computer Science and Engineering**

CSE471: Computer and Network Security

Final Examination

Semester: Spring 2025

Batch: 23rd

Time: 2 Hours

Date: 10/03/26

Answer any **five questions** from the followings.

**Total Marks: 40**

**1.** a) "An Elliptic Curve Cryptography (ECC) provides equivalent or higher security than RSA while using much smaller key sizes, making it more efficient in terms of computation and storage." – justify with your answer. **3**

b) Show the complete key establishment and encryption and decryption process of a message using Elliptic Curve Cryptography (ECC). **5**

**2.** a) What is Advanced Encryption Standard (AES)? **2**

b) State the differences between stream cipher and block cipher. **3**

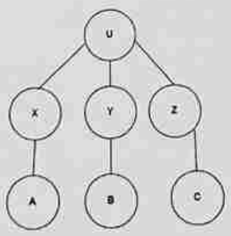
c) Why is 3DES more secure than DES? Explain. **3**

**3.** a) What is the primary purpose of the Oakley Key Determination Protocol and how does it enhance Diffie-Hellman key exchange? **4**

b) What is SSL and what is its primary purpose? Explain the SSL handshaking protocol and illustrate the key steps involved in establishing a secure connection. **4**

**4.** a) What is a digital certificate? Explain. **4**

b) In an X.509 authentication framework, entities U, X, Y, Z, A, B, and C follow the Chain of Certificates rule. Given the certificate hierarchy as shown, explain how entity A can securely communicate with entity C? **4**




**5.** a) Explain the use of a rainbow table in Birthday Attack. **2**

b) Explain how PGP (Pretty Good Privacy) works, including its cryptographic processes. **6**
Provide 3 individual scenarios where PGP is used and show whether they provide authentication, confidentiality or both. Include necessary diagrams or illustrations to support your explanation.

**6.** a) Explain the role of cryptography in the UNIX password scheme, including how cryptographic techniques are used to securely store and verify user passwords. **4**